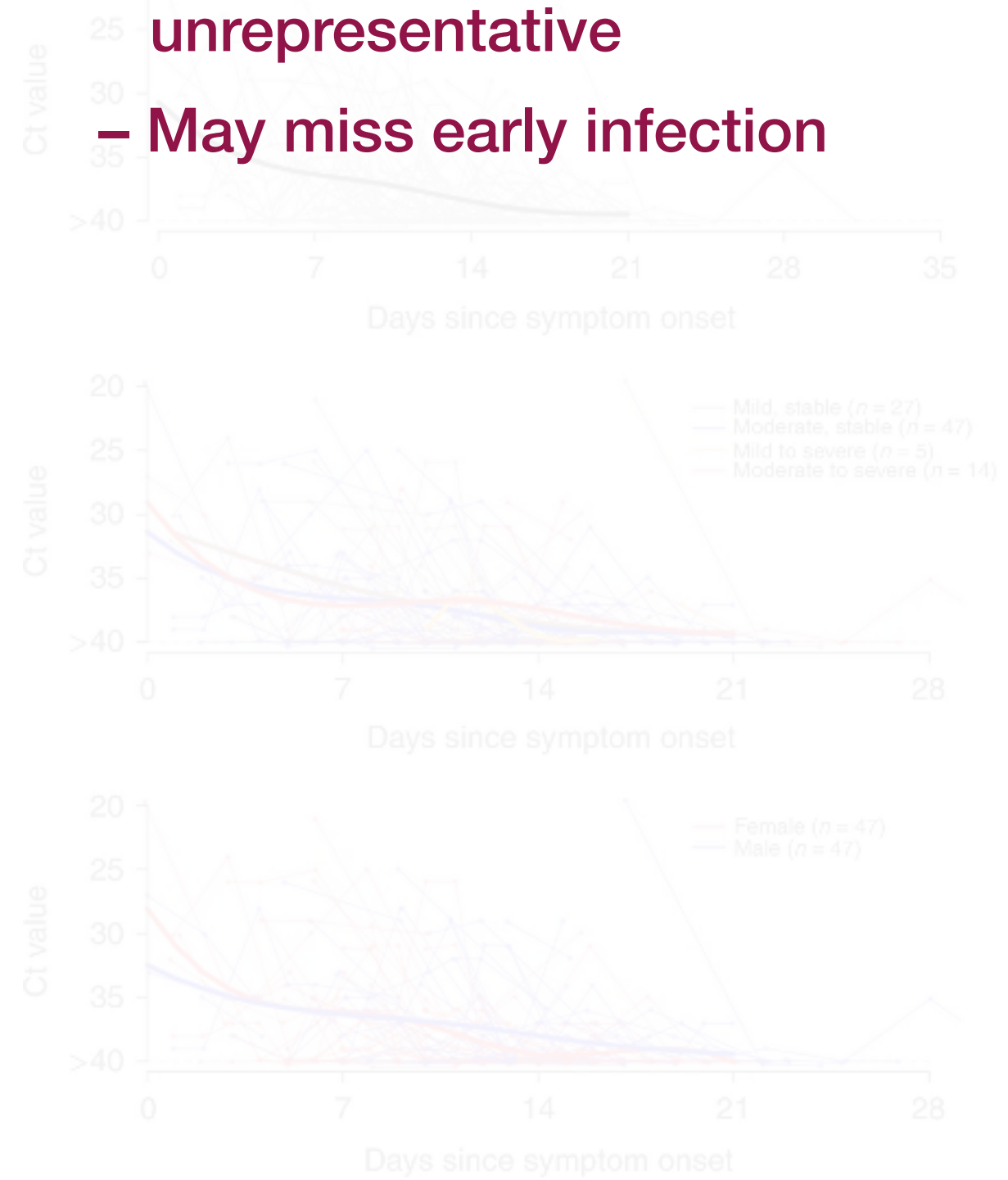


Viral kinetics data during the COVID-19 pandemic

Targeted studies

- + Quick to design/deploy
- + Can be highly informative about key epi parameters

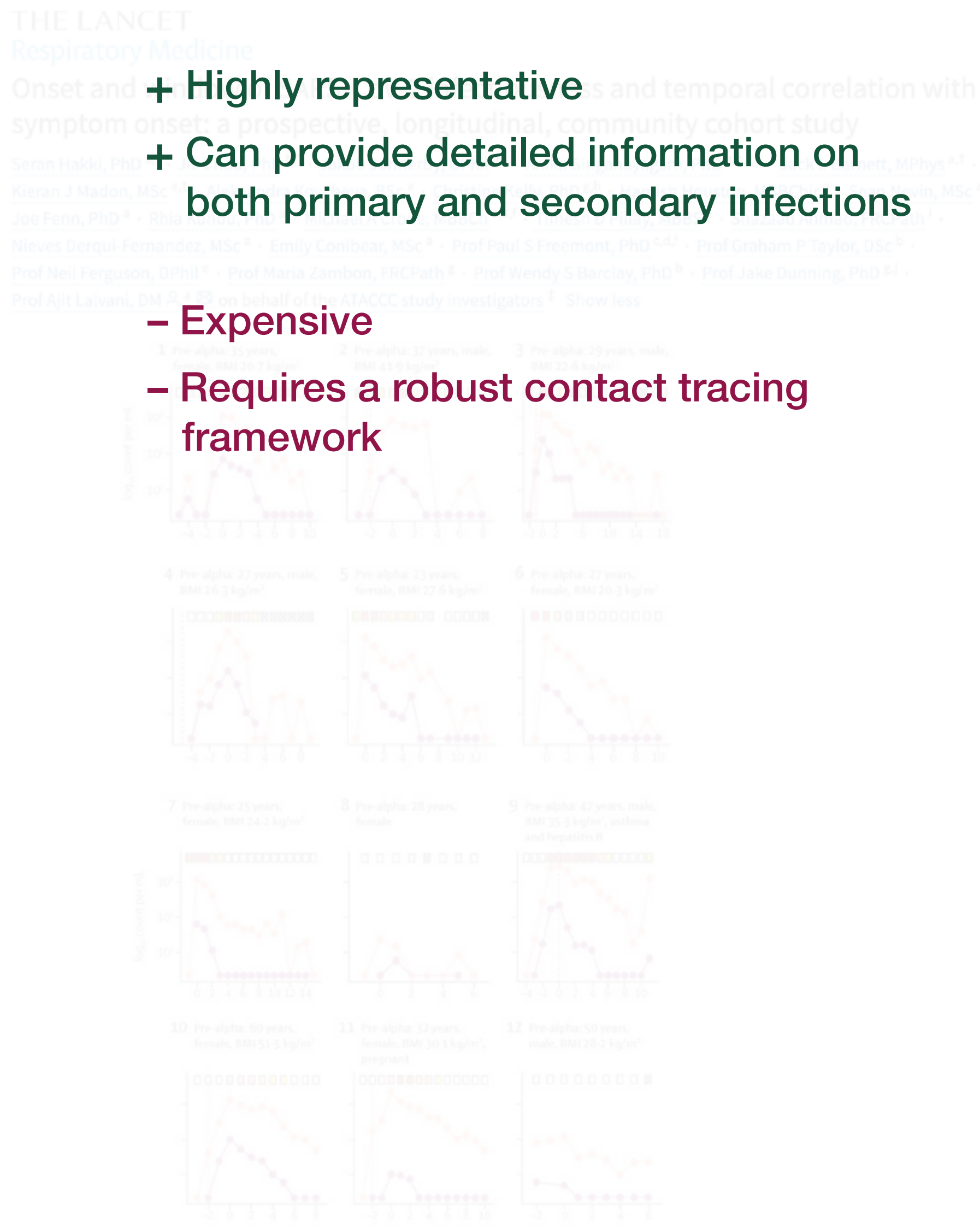
- Sample sizes often small, unrepresentative
- May miss early infection



Large prospective studies

- + Highly representative
- + Can provide detailed information on both primary and secondary infections

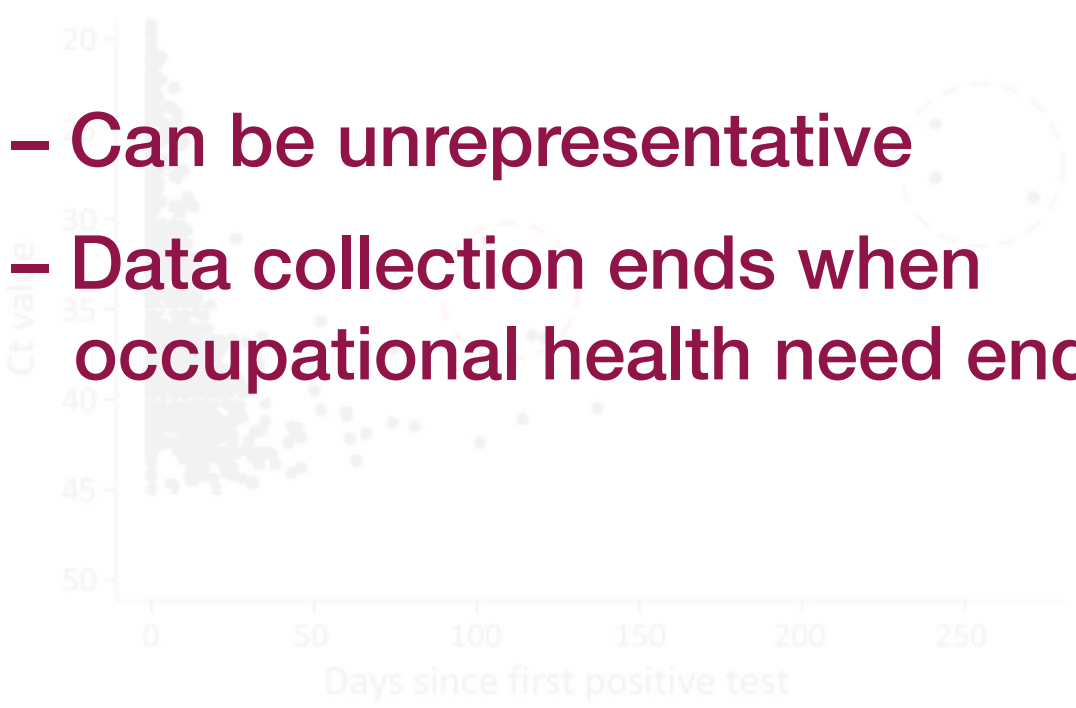
- Expensive
- Requires a robust contact tracing framework



Occupational health programs

- + Can generate huge volumes of data
- + “Cheap” in the sense that the data’s already being collected

- Can be unrepresentative
- Data collection ends when occupational health need ends



Viral kinetics of sequential SARS-CoV-2 infections

Stephen M. Kissler, James A. Haw, Joseph B. Faauer, Christina Mack, Caroline D. Tai, Derrick J. Anderson, David D. Ho, Nathan D. Grubaugh & Yonatan H. Grad

Nature Communications 14, Article number: 6206 (2023) | Cite this article



A proposal: integrating viral kinetics into routine surveillance

A continuous monitoring system for the kinetics of endemic and emerging pathogens...

...that would complement active surveillance efforts (e.g., hospitalizations, mortality, serology, genomics)...

...and integrate into dashboards and situation reports alongside estimates of prevalence and similar metrics.

